Case Records of the Wexford General Hospital

A Clinicopathological Exercise • Synthetic Teaching Case

Case 35-2026: A 67-Year-Old Man with Progressive Heart Failure, Low-Voltage Electrocardiogram, and Nephrotic-Range Proteinuria

The following is an entirely fictional teaching case. All names, dates, case numbers, institutions, laboratory values, and narrative details are invented for instructional purposes and do not depict any real patient, physician, or institution.

Presentation of the Case

Dr. Caleb Mensah (Medicine): A 67-year-old man was admitted to this hospital because of eight months of progressive exertional dyspnea, fatigue, and lower-extremity swelling.

The patient had been well until approximately eight months before admission, when he first noticed breathlessness on climbing the single flight of stairs to his bedroom, having previously walked two miles on level ground without limitation. The dyspnea advanced steadily, so that by admission he was short of breath after crossing a room, and he had begun to sleep propped on three pillows to avoid waking with a sensation of suffocation. Swelling of both ankles progressed over weeks to involve the calves, thighs, and scrotum, and his abdomen had become distended despite his eating less.

During the same interval he had profound fatigue and episodes of lightheadedness on standing, twice nearly falling when he rose quickly, such that he had learned to stand slowly. His appetite had changed markedly: he felt uncomfortably full after a few mouthfuls and had lost an estimated 9 kg without intending to. He denied chest pain, palpitations, and syncope but reported a dull right-sided abdominal fullness and increasing constipation.

On focused questioning, additional symptoms emerged. For roughly a year he had noticed tingling and numbness of both feet that had slowly ascended to the mid-shin. He bruised very easily, and he and his wife had noticed dark discoloration around both eyes that appeared after coughing or straining. His voice had become soft and hoarse, and his urine appeared foamy.

His medical history was notable for bilateral carpal tunnel syndrome, treated with surgical release of both wrists approximately two years before admission. He had carried a diagnosis of "borderline" hypertension for many years, managed with a single agent; notably, over the preceding six to nine months his home readings had fallen progressively, and his physician had first reduced and then stopped the antihypertensive because of low readings and lightheadedness. He had never had a myocardial infarction, atrial fibrillation, diabetes mellitus, or known kidney disease, and there was no history of rheumatologic disease or chronic inflammation.

His only regular medication on admission was furosemide, 40 mg daily, started six weeks earlier for the leg swelling; amlodipine and lisinopril, taken for years, had both been stopped in the preceding months. He took no anticoagulants or nonsteroidal agents and had no known drug allergies. His immunizations were current.

The patient was a retired carpenter and cabinetmaker who lived with his wife. He had smoked half a pack of cigarettes daily in his twenties and thirties but had stopped more than thirty years earlier, drank one beer on weekends, and had never used illicit drugs. He reported no foreign travel beyond a single trip to Canada decades earlier and no unusual occupational or animal exposures. His family history was carefully reviewed and was unrevealing for cardiomyopathy, sudden cardiac death, heart failure at a young age, hereditary neuropathy, or kidney disease; his parents had lived into their eighties.

On review of systems, he confirmed exertional dyspnea, orthopnea, dependent edema, postural lightheadedness, early satiety, constipation, distal paresthesias, easy bruising, periorbital discoloration, hoarseness, and foamy urine. He denied fever, night sweats, cough, hemoptysis, arthralgias, rash other than the bruising described, and sicca symptoms.

On examination, the patient was a thin, tired-appearing man in no acute distress at rest but becoming dyspneic with minimal exertion. The temperature was 36.7°C, the heart rate 78 beats per minute and regular, the respiratory rate 16 breaths per minute, and the oxygen saturation 95% while he was breathing ambient air. The blood pressure was 106/74 mm Hg while he was supine; on standing, it fell to 84/58 mm Hg within two minutes, accompanied by lightheadedness, without an appropriate compensatory rise in heart rate (the pulse increased by only 6 beats per minute). The jugular venous pressure was markedly elevated, to approximately 13 cm of water, with a prominent and rapid descent in the venous waveform, and it failed to fall — indeed appeared to rise — with inspiration. Carotid upstrokes were normal in volume and timing.

Cardiac examination revealed a quiet precordium with a nondisplaced apical impulse, a soft first and second heart sound, and an early third heart sound (S3) at the apex; there was a soft, grade 2/6 systolic murmur at the lower left sternal border that increased slightly with inspiration. The lungs had fine inspiratory crackles at both bases. The abdomen was distended, with shifting dullness consistent with ascites; the liver was enlarged and firm, palpable approximately 4 cm below the right costal margin, with a smooth edge that seemed to pulsate with the venous pulse. The spleen was not palpable. There was pitting edema extending from the feet to the thighs, with scrotal edema.

Examination of the skin and soft tissues disclosed several findings. There was well-demarcated purpura of both upper eyelids and the periorbital skin — so-called "raccoon eyes" — together with scattered ecchymoses of varying ages on the forearms. The tongue was enlarged, with scalloped indentations along its lateral borders corresponding to the teeth, and it appeared firm; the submandibular regions were full. The voice was soft and slightly hoarse. There was no lymphadenopathy in the cervical, axillary, or inguinal regions.

On neurologic examination, he was alert and fully oriented, with normal cranial nerves and full strength. There was symmetric reduction of pinprick and light-touch sensation in a stocking distribution to the midshin, markedly reduced vibration at the great toes, and absent ankle reflexes, with flexor plantar responses. Gait was cautious, with difficulty on tandem testing.

The patient was admitted for treatment of decompensated heart failure. Intravenous diuresis was begun but was repeatedly limited by symptomatic orthostatic hypotension that worsened as he was decongested. Laboratory studies are shown in **Table 1**, **Table 2**, **Table 3**, and **Table 4**.

Table 1. Hematologic and Coagulation Studies.

Variable	Patient's Value	Reference Range
Hemoglobin (g/dL)	13.1	13.5–17.5
Hematocrit (%)	39.4	41.0–53.0
Mean corpuscular volume (fL)	89	80–100
White-cell count (×10⁹/L)	6.4	4.5–11.0
Platelet count (×10⁹/L)	232	150–400
Reticulocyte count (%)	1.4	0.5–2.5
Peripheral smear	Normochromic, normocytic indices; rare rouleaux; no schistocytes; no blasts	—
Prothrombin time (s)	16.2	11.5–14.5
International normalized ratio	1.4	0.9–1.1
Activated partial-thromboplastin time (s)	33	25–35
Fibrinogen (mg/dL)	332	200–400

Table 2. Clinical Chemistry and Hepatic Studies.

Variable	Patient's Value	Reference Range
Sodium (mmol/L)	136	135–145
Potassium (mmol/L)	4.5	3.5–5.0
Chloride (mmol/L)	101	98–107
Carbon dioxide (mmol/L)	25	22–30
Urea nitrogen (mg/dL)	29	8–20
Creatinine (mg/dL)	1.4	0.7–1.3
Estimated GFR (mL/min/1.73 m²)	52	>60
Glucose (mg/dL)	92	70–100
Calcium (mg/dL)	9.1	8.5–10.2
Total protein (g/dL)	5.6	6.0–8.3
Albumin (g/dL)	2.9	3.5–5.0
Aspartate aminotransferase (U/L)	31	10–40
Alanine aminotransferase (U/L)	26	7–52
Alkaline phosphatase (U/L)	196	40–129
Gamma-glutamyl transferase (U/L)	118	8–61
Total bilirubin (mg/dL)	0.8	0.3–1.2
Lactate dehydrogenase (U/L)	188	110–210
Thyrotropin (μIU/mL)	2.1	0.4–4.0

Table 3. Cardiac Biomarkers.

Variable	Patient's Value	Reference Range
N-terminal pro–B-type natriuretic peptide (pg/mL)	6,420	<125
High-sensitivity cardiac troponin T (ng/L)	62	<14
Creatine kinase (U/L)	88	30–200

Table 4. Renal and Monoclonal-Protein Studies.

Variable	Patient's Value	Reference Range
Urinalysis, protein	3+	Negative
Urinalysis, blood	Negative	Negative
Urinalysis, sediment	Bland; no cellular casts; oval fat bodies present	—
24-hour urine total protein (g/24 h)	6.2	<0.15
24-hour urine albumin (% of total protein)	~80	—
Serum free kappa light chain (mg/L)	11.2	3.3–19.4
Serum free lambda light chain (mg/L)	240	5.7–26.3
Serum free kappa:lambda ratio	0.05	0.26–1.65
Difference between involved and uninvolved free light chain, dFLC (mg/L)	229	<18
Serum protein electrophoresis	No discrete M-spike on densitometry; hypoalbuminemia	—
Serum immunofixation	Monoclonal free lambda light chain detected	—
Urine immunofixation	Monoclonal free lambda light chain detected	—
Beta-2-microglobulin (mg/L)	3.6	0.7–1.8
Immunoglobulin G / A / M	Within or below normal limits	—

Serologic testing for human immunodeficiency virus, hepatitis B, and hepatitis C was negative. Antinuclear antibody, antineutrophil cytoplasmic antibodies, rheumatoid factor, and serum and urine cultures were unremarkable or negative. The C-reactive protein was 4 mg/L (reference range, <5), and the erythrocyte sedimentation rate was 22 mm per hour. Coronary evaluation, undertaken because of the elevated troponin, revealed no obstructive epicardial coronary disease and no regional wall-motion abnormality to suggest prior infarction.

Dr. Mensah: A 12-lead electrocardiogram showed normal sinus rhythm at a rate of 74 beats per minute with first-degree atrioventricular block (PR interval, 224 ms). The most striking feature was strikingly low QRS voltage in the limb leads, with the QRS amplitude in each of the six limb leads measuring 5 mm or less. There was poor R-wave progression across the precordial leads, with diminutive R waves and small Q waves in leads V1 through V3, producing a "pseudoinfarct" pattern in the anteroseptal distribution in a patient with no history or angiographic evidence of myocardial infarction. There was no left-ventricular hypertrophy by voltage criteria despite — as will be seen — substantially increased wall thickness on imaging. The QT interval was normal, and there were no acute ST-segment or T-wave changes.

Transthoracic echocardiography was performed. The left ventricle was normal in cavity size but showed concentric thickening of the walls, with an interventricular septal thickness of 16 mm and a posterior-wall thickness of 15 mm. The myocardium had an unusually bright, granular, "sparkling" texture. There was marked biatrial enlargement, thickening of the interatrial septum, and diffuse thickening of the mitral, tricuspid, and aortic valve leaflets without significant stenosis. A small circumferential pericardial effusion was present, without tamponade physiology. Right-ventricular free-wall thickness was increased. Doppler interrogation demonstrated a restrictive left-ventricular filling pattern, with a tall, narrow early (E) filling wave, a small atrial (A) wave, a short deceleration time, and a markedly elevated ratio of early mitral inflow velocity to early diastolic mitral-annular velocity (E/e′), indicating high filling pressures. The left-ventricular ejection fraction was 50 percent. Speckle-tracking analysis revealed severely reduced global longitudinal strain with a characteristic pattern of relative preservation of strain at the left-ventricular apex and severe impairment of the basal and mid-ventricular segments — the so-called apical-sparing or "cherry-on-top" bullseye pattern.

Dr. Priya Raman (Radiology): Cardiac magnetic resonance imaging confirmed the concentric increase in left- and right-ventricular wall thickness and biatrial enlargement seen on echocardiography. After the administration of gadolinium contrast material, there was diffuse, predominantly subendocardial late gadolinium enhancement extending circumferentially around both ventricles and involving the atrial walls and the interatrial septum. The myocardial and blood-pool nulling behaved abnormally on the inversion-recovery sequences, such that the myocardium and blood pool nulled at similar inversion times — a recognized signature of abnormal gadolinium kinetics in infiltrative myocardial disease. Parametric mapping showed markedly elevated native myocardial T1 and a greatly expanded extracellular volume fraction, indicating expansion of the myocardial interstitium by an infiltrative material rather than by replacement fibrosis alone.

Technetium-99m pyrophosphate (PYP) scintigraphy was performed. On planar and single-photon-emission computed tomographic imaging at one and three hours, there was only faint myocardial radiotracer uptake, graded as 0 to 1 (uptake less than or equal to that of the ribs), with a heart-to-contralateral-lung ratio below the diagnostic threshold. Blood-pool activity was appropriately cleared. The degree of myocardial uptake was therefore not in the range considered diagnostic of transthyretin cardiac amyloidosis.

Dr. Mensah: In summary, this was a 67-year-old man with a subacute, multisystem illness centered on a restrictive cardiomyopathy with preserved ejection fraction, accompanied by nephrotic-range proteinuria, a sensory and autonomic peripheral neuropathy, hepatomegaly with cholestatic liver-enzyme abnormalities, distinctive soft-tissue findings, and a monoclonal free lambda light chain in serum and urine. A diagnostic procedure was performed.

Differential Diagnosis

Dr. Eleanor Whitfield (Cardiology): The central interpretive principle in this case is that although the patient was admitted for heart failure, this is not an isolated cardiac illness. He has a *systemic* disease that declares itself most prominently in the heart. The diagnostic task is to account for every organ that is involved.

The systems involved can be enumerated. There is a *cardiac* syndrome: progressive heart failure with a restrictive filling pattern, a preserved ejection fraction, thick ventricular walls, biatrial enlargement, valvular and septal thickening, a pericardial effusion, low electrocardiographic voltage, a markedly elevated natriuretic peptide, and a persistently elevated troponin without coronary disease. There is a *renal* syndrome: nephrotic-range proteinuria of 6.2 g per day that is predominantly albumin, with hypoalbuminemia, edema, and a mildly reduced glomerular filtration rate. There is a *neurologic* syndrome: a length-dependent sensory peripheral neuropathy together with prominent autonomic failure, manifested by orthostatic hypotension without compensatory tachycardia, constipation, and early satiety. There is a *hepatic* syndrome: hepatomegaly with a cholestatic pattern of liver-enzyme elevation — a high alkaline phosphatase and gamma-glutamyl transferase with preserved transaminases and bilirubin. There is also a striking constellation of *soft-tissue and hemostatic* findings: macroglossia with dental scalloping, periorbital purpura, easy bruising, a prolonged prothrombin time, and hoarseness. Finally, two antecedent clues are woven through the history: bilateral carpal tunnel syndrome requiring surgical release two years earlier, and the resolution of long-standing hypertension with withdrawal of antihypertensive drugs.

A diagnosis that explains heart failure alone is insufficient. The unifying diagnosis must simultaneously involve the heart, kidney, nerves, liver, and soft tissues. The differential diagnosis can begin with the cardiac physiology, because correctly classifying the cardiomyopathy narrows the field substantially.

Defining the physiology: a restrictive cardiomyopathy with "pseudohypertrophy"

The echocardiographic and hemodynamic picture is that of a restrictive cardiomyopathy: a non-dilated ventricle with thick walls, severe diastolic dysfunction with a restrictive filling pattern, biatrial enlargement reflecting chronically elevated atrial pressures, and an ejection fraction that is preserved or only mildly reduced even as the patient is profoundly symptomatic. The elevated jugular venous pressure with a rapid descent, the failure of the venous pressure to fall with inspiration, the S3, and the systemic congestion all corroborate a stiff, non-compliant ventricle.

The most discriminating observation in the case is the *discordance between wall thickness and electrocardiographic voltage*. The echocardiogram shows walls of 15 to 16 mm — well above normal — yet the electrocardiogram shows low limb-lead voltage and no hypertrophy by any voltage criterion. In ordinary causes of a thick-walled ventricle, increased muscle mass produces *increased* QRS voltage. When the walls are thick but the voltage is low, the thickening is not true myocyte hypertrophy but rather *infiltration* of the myocardial interstitium by a substance that is electrically inert and that pushes the walls apart while the population of functioning myocytes shrinks. This "pseudohypertrophy" with low voltage is the cardinal clue of an infiltrative cardiomyopathy, and it reorients the differential away from the common causes of left-ventricular hypertrophy.

Two further imaging findings deserve emphasis because they are nearly specific. First, the apical-sparing or "cherry-on-top" pattern of longitudinal strain — severe impairment basally and at the mid-ventricle with relative preservation at the apex — is a well-recognized fingerprint of cardiac amyloidosis and helps distinguish it from other causes of a thick-walled ventricle. Second, the cardiac magnetic resonance findings — diffuse subendocardial-to-transmural late gadolinium enhancement, abnormal myocardial nulling with co-nulling of blood and myocardium, and a markedly expanded extracellular volume — together describe expansion of the interstitium by an infiltrate, not the patchy mid-wall scar of other cardiomyopathies. The biomarker profile fits as well: a natriuretic peptide above 6,000 pg/mL and a troponin persistently elevated to 62 ng/L in the absence of coronary disease indicate ongoing myocyte stress and injury from infiltration rather than ischemia.

Distinguishing restriction from constriction

Restrictive cardiomyopathy must be distinguished from constrictive pericarditis. Both produce elevated jugular venous pressure with rapid descents, congestion, ascites, edema, and a restrictive filling pattern, and the distinction matters because constriction is surgically curable by pericardiectomy. Several features here argue decisively for restriction from myocardial disease rather than constriction. The walls are genuinely thickened and the myocardium is abnormal on every modality; constriction typically shows normal myocardium with a thickened or calcified pericardium and ventricular interdependence with respirophasic septal shift, none of which is de-

scribed. The natriuretic peptide is very high; in pure constriction, where the myocardium is relatively spared, natriuretic peptides tend to be only modestly elevated, whereas in restrictive myocardial disease they are markedly elevated, as here. Biatrial enlargement of this degree, severe abnormalities of longitudinal strain, and infiltrative enhancement on magnetic resonance imaging all point to the muscle, not the sac. The findings therefore indicate a restrictive cardiomyopathy of myocardial origin.

The thick-walled ventricle with low voltage: excluding the common mimics

The common causes of a thick left ventricle fail to fit either the electrocardiogram or the multisystem picture.

Hypertensive heart disease would be the reflexive explanation in a man with a history of hypertension. But it does not fit, and in fact the history actively argues against it. Hypertensive hypertrophy produces *high* QRS voltage, not low voltage; it does not cause biatrial enlargement of this degree, valvular and septal thickening, a pericardial effusion, an infiltrative enhancement pattern, or — most importantly — nephrotic-range proteinuria, neuropathy, macroglossia, and a monoclonal protein. And there is the remarkable clue that his blood pressure *fell* over the past year to the point that his antihypertensives had to be stopped. When long-standing hypertension melts away spontaneously and the patient becomes orthostatic, one should suspect that the pump is failing and that the autonomic nervous system is being destroyed — both of which an infiltrative systemic disease accomplishes. The de-escalation of his antihypertensive therapy is not incidental; it is a symptom.

Hypertrophic cardiomyopathy can produce marked wall thickening, but it characteristically shows high voltage, an asymmetric septal pattern often with systolic anterior motion of the mitral valve and outflow obstruction, and it presents earlier in life, frequently with a family history of the disease or of sudden death. This man has concentric thickening, low voltage, no obstruction, late onset, and no family history, and hypertrophic cardiomyopathy does not cause proteinuria, neuropathy, or macroglossia.

Severe aortic stenosis causes pressure-overload hypertrophy, but the valves here, although thickened, are not stenotic, and aortic stenosis would not explain the systemic findings.

The low-voltage, infiltrative phenotype directs attention to the storage and infiltrative cardiomyopathies.

Storage and infiltrative cardiomyopathies

Fabry disease, an X-linked deficiency of alpha-galactosidase A, deserves genuine consideration, because it causes concentric left-ventricular thickening, can be associated with relatively low voltage, and is a multisystem disease that affects the kidney (proteinuria and progressive renal failure) and the peripheral nerves (a painful small-fiber neuropathy, the acroparesthesias). However, Fabry disease typically declares itself far earlier in life, often with childhood or adolescent acroparesthesias, angiokeratomas, hypohidrosis, and the characteristic corneal opacity (cornea verticillata), and it is usually accompanied by a family history consistent with X-linked inheritance. Its cardiac magnetic resonance signature is classically a basal inferolateral mid-wall late-enhancement pattern with a *low* native T1 (because of intracellular sphingolipid), not the diffuse subendocardial enhancement with markedly *elevated* T1 and expanded extracellular volume seen here. Most decisively, Fabry disease does not cause macroglossia, periorbital purpura, a monoclonal light chain, or a coagulopathy. An alpha-galactosidase A level would be appropriate to check, but the overall phenotype is not consistent with Fabry disease.

Cardiac sarcoidosis is the other great infiltrative imitator. It can cause conduction disease — and this patient does have first-degree atrioventricular block — and it produces patchy myocardial enhancement. But sarcoid cardiomyopathy typically causes *regional* wall-motion abnormalities, aneurysms, and patchy, often basal-septal or mid-wall, late enhancement rather than the diffuse, concentric, subendocardial infiltration documented here; it tends to present with high-grade heart block or ventricular arrhythmias; and it does not cause nephrotic-range albuminuria, macroglossia, periorbital purpura, or a monoclonal light chain. The normal inflammatory markers and the absence of pulmonary, ocular, or cutaneous sarcoid further argue against it.

Cardiac hemochromatosis (iron-overload cardiomyopathy) usually produces a dilated, rather than a thick, restrictive ventricle, is identified by a low myocardial T2 *on magnetic resonance imaging, and is accompanied by abnormal iron studies and often by diabetes and hepatic disease of a different character; it does not fit. The* glycogen-storage and lysosomal cardiomyopathies* (Danon disease, PRKAG2 cardiomyopathy, Pompe disease)

present early in life, usually with very high voltage and pre-excitation, and are excluded by age and phenotype.

Each of these infiltrative diseases can take the heart, and one or two can take the kidney or the nerve. But none of them ties together the heart, the kidney, the nerves, the liver, the tongue, the periorbital skin, the bleeding tendency, *and* a monoclonal protein. There is only one family of disease that does.

Amyloidosis: the unifying diagnosis

Amyloidosis is the deposition, in the extracellular space, of insoluble protein fibrils with a characteristic beta-pleated-sheet structure. It is an infiltrative, multisystem disease, and it accounts for the elements of this case more economically than the competing diagnoses.

The *cardiac* picture is classic amyloid cardiomyopathy: amyloid fibrils deposited throughout the myocardial interstitium thicken the walls while replacing and surrounding functioning myocytes, producing the thick-walled, low-voltage "pseudohypertrophy" seen here; the granular sparkling texture, biatrial enlargement, valvular and interatrial-septal thickening, pericardial effusion, restrictive filling, apical-sparing strain, diffuse subendocardial late enhancement, elevated native T1, and expanded extracellular volume are characteristic imaging signatures of amyloid infiltration. The persistently elevated troponin reflects ongoing myocyte injury from the infiltrate, and the natriuretic peptide of 6,420 pg/mL reflects the wall stress of a stiff, failing ventricle. The pseudoinfarct pattern and the first-degree atrioventricular block reflect amyloid infiltration of the conduction system and loss of anterior myocardial forces.

The *renal* picture — nephrotic-range proteinuria that is predominantly albumin, with hypoalbuminemia and edema — is the signature of glomerular amyloid deposition, which damages the filtration barrier and produces heavy albuminuria, in contrast to the tubular, light-chain proteinuria of myeloma cast nephropathy. The oval fat bodies and the bland sediment fit a nephrotic process without nephritis.

The *neurologic* picture — a length-dependent sensory neuropathy with prominent autonomic failure — is amyloid polyneuropathy and autonomic neuropathy. The orthostatic hypotension without compensatory tachycardia is the fingerprint of autonomic denervation, and it, together with the failing pump, explains both his lightheadedness and the disappearance of his hypertension. The early satiety and constipation reflect autonomic gastrointestinal dysmotility, and contribute to his weight loss.

The *hepatic* picture — hepatomegaly with a cholestatic enzyme pattern (a high alkaline phosphatase and gamma-glutamyl transferase with preserved bilirubin and transaminases) — is characteristic of hepatic amyloid infiltration. The firm, apparently pulsatile liver also reflects the transmitted venous pulsation of his elevated right-sided pressures.

The *soft-tissue and hemostatic* findings are, frankly, the clincher. *Macroglossia* with lateral scalloping from the teeth is a classic and rather specific sign of amyloidosis. *Periorbital purpura* — "raccoon eyes" appearing after coughing or straining — results from amyloid infiltration of dermal and small-vessel walls causing capillary fragility, and is one of the most evocative physical signs in all of medicine for this disease. The easy bruising and the prolonged prothrombin time additionally raise the possibility of acquired factor X deficiency, which occurs in amyloidosis because factor X is adsorbed onto amyloid fibrils, particularly in the liver and spleen. The hoarseness may reflect amyloid deposition in the larynx or soft tissues.

And then there are the two antecedent clues. *Bilateral carpal tunnel syndrome* preceding the cardiac presentation by two years is a well-described herald of systemic amyloidosis: amyloid deposits in the tenosynovium of the flexor retinaculum compress the median nerve, often years before the heart declares itself. Its bilateral nature and its appearance in a man in his sixties should, in retrospect, have prompted consideration of amyloid even then. This single historical fact, taken with everything else, is powerful.

The unifying diagnosis, therefore, is *systemic amyloidosis with cardiac, renal, hepatic, neural, and soft-tissue involvement*. But naming "amyloidosis" is only half of the diagnosis. The treatment, the prognosis, and the clinical meaning of the disease depend entirely on *which protein* is forming the fibrils. The amyloid must therefore be typed.

Typing the amyloid: AL versus ATTR versus AA

There are several systemic amyloidoses, and three are common enough and compatible enough with this picture to require explicit consideration: AL (immunoglobulin

light-chain) amyloidosis, ATTR (transthyretin) amyloidosis in both its hereditary and wild-type forms, and AA (serum amyloid A) amyloidosis.

AA amyloidosis is a complication of chronic inflammatory disease — long-standing rheumatoid arthritis, inflammatory bowel disease, chronic infection, or a hereditary periodic fever syndrome — in which the acute-phase reactant serum amyloid A is chronically overproduced and deposits as fibrils. It is predominantly a renal disease, presenting with proteinuria, and it can involve the liver and gut. But this patient has no chronic inflammatory illness, his C-reactive protein and sedimentation rate are essentially normal, and AA amyloidosis does not characteristically cause macroglossia, periorbital purpura, the acquired coagulopathy, or this degree of infiltrative cardiomyopathy. AA amyloidosis is most unlikely, and the presence of a monoclonal light chain points away from it.

ATTR amyloidosis is, in many respects, the most important alternative to exclude, and it shares several features with this case. Wild-type ATTR (formerly "senile systemic amyloidosis") characteristically affects older men, causes an infiltrative cardiomyopathy with the same low-voltage, thick-walled, apical-sparing phenotype, and is famously preceded by bilateral carpal tunnel syndrome and by lumbar spinal stenosis and biceps-tendon rupture. Hereditary ATTR, caused by pathogenic variants in the transthyretin gene, can cause a combined cardiomyopathy and polyneuropathy with autonomic features and can occur without a clear family history if penetrance is low or the family history is incomplete. So a thoughtful clinician confronted with an older man who has carpal tunnel syndrome, an infiltrative cardiomyopathy, and a neuropathy *must* consider ATTR. Indeed, the carpal tunnel history, which I cited as a clue to amyloid, points to both ATTR and AL and does not by itself discriminate between them.

Several features, however, favor AL over ATTR in this patient. First, the soft-tissue findings — macroglossia and periorbital purpura — are hallmarks of AL amyloidosis and are distinctly uncommon in ATTR; the same is true of the acquired factor-X coagulopathy. Second, ATTR amyloidosis rarely causes heavy, nephrotic-range glomerular proteinuria; significant renal involvement of this magnitude is far more typical of AL. Third, and importantly, the technetium-99m pyrophosphate scan showed only grade 0–1 myocardial uptake. Intense myocardial pyrophosphate uptake (grade 2–3, with a heart-to-contralateral ratio at or above 1.5) is highly sensitive for ATTR cardiac amyloidosis; the absence of significant uptake here makes ATTR unlikely and is consistent with a non-ATTR — that is, an AL — process. And fourth, the patient has an unequivocal clonal abnormality: a monoclonal free lambda light chain in serum and urine by immunofixation, a serum free lambda level of 240 mg/L, and a profoundly suppressed kappa:lambda ratio of 0.05, with a dFLC of 229 mg/L. That is the biochemical signature of a lambda-restricted plasma-cell clone producing an amyloidogenic light chain.

A subtle but essential point of reasoning follows. The presence of a monoclonal protein does *not*, by itself, prove that the amyloid is of AL type. Monoclonal gammopathy of undetermined significance is common in older adults — present in a substantial minority of people over 70 — so a patient can have wild-type ATTR cardiac amyloidosis *and* an unrelated, coincidental monoclonal protein. If AL were assumed on the basis of the monoclonal protein alone and the amyloid were in fact ATTR, the patient would be exposed to toxic and useless chemotherapy while the true disease went untreated. For exactly this reason, the diagnosis of AL amyloidosis cannot rest on the serum studies; it requires *tissue confirmation of amyloid and definitive typing of the amyloid protein*. That said, in this patient the convergence is compelling: a markedly abnormal light-chain profile with a high dFLC and strongly skewed ratio, a negative pyrophosphate scan arguing against ATTR, and the AL-specific soft-tissue and hemostatic findings. The clinical probability of AL amyloidosis is very high.

The magnitude of the free lambda elevation, the suppression of the uninvolved kappa, and the large dFLC also tell me that the responsible clone is not trivial; combined with the rouleaux on the smear and the borderline-low immunoglobulins, this suggests a clonal plasma-cell population in the marrow, which must be quantified.

The diagnostic procedure

The indicated procedure is *tissue sampling to demonstrate amyloid by Congo red staining, followed by definitive typing of the amyloid protein*, together with a bone marrow biopsy to characterize and quantify the plasma-cell clone. The least invasive and highest-yield first step is an *abdominal subcutaneous fat-pad aspiration or biopsy*, which is positive in a large majority of patients with systemic AL amyloidosis and carries minimal morbidity; a bone marrow biopsy obtained at the

same time both can be stained for amyloid and provides the clonal plasma-cell assessment. If the fat pad and marrow were unrevealing in the face of this strong clinical suspicion, then biopsy of an involved organ would be warranted — and *endomyocardial biopsy* is the definitive cardiac test, approaching 100 percent sensitivity for cardiac amyloid, though it is usually unnecessary when extracardiac tissue confirms the diagnosis. Crucially, whatever tissue is obtained must not only be stained with Congo red but must be *typed* — ideally by mass spectrometry — to prove that the fibrils are composed of immunoglobulin lambda light chain and to exclude transthyretin and serum amyloid A.

My leading diagnosis is systemic AL (lambda light-chain) amyloidosis with cardiac, renal, hepatic, neurologic, and soft-tissue involvement.

Clinical Diagnosis

Systemic light-chain (AL) amyloidosis with infiltrative restrictive cardiomyopathy, nephrotic-range proteinuria, hepatic and autonomic involvement, and a lambda-restricted plasma-cell clone.

Dr. Eleanor Whitfield's Diagnosis

Systemic AL (lambda immunoglobulin light-chain) amyloidosis with cardiac involvement.

Pathological Discussion

Dr. Tomás Herrera (Pathology): The diagnostic procedure comprised an abdominal subcutaneous fat-pad aspiration together with a bone marrow aspirate and core biopsy. Both established the diagnosis, and the case illustrates the modern, multistep approach to confirming amyloidosis and — equally important — to determining its type.

The fat-pad aspirate was smeared, fixed, and stained with Congo red. Under ordinary light microscopy, the sections contained amorphous, waxy, eosinophilic extracellular material deposited around small vessels and within the interstitium of the adipose tissue. With the Congo red stain, this material took up the dye and appeared salmon-pink under standard transmitted light. When the stained section was then examined under cross-polarized light, the deposits exhibited the pathognomonic *apple-green birefringence* — a shift of the polarization color to a brilliant green — that defines amyloid and reflects the ordered, beta-pleated-sheet architecture of the fibrils binding the Congo red dye in register. This birefringence under polarized light is the gold-standard histochemical demonstration of amyloid and distinguishes true amyloid from other eosinophilic hyaline materials.

The bone marrow core biopsy and aspirate were diagnostic of the underlying clonal process. The marrow was normocellular for the patient's age, but among the hematopoietic elements there was an increased population of plasma cells, estimated at 10 to 12 percent of marrow cellularity, with mild nuclear atypia and occasional binucleate forms. Immunohistochemical staining for kappa and lambda light chains, corroborated by in situ hybridization for light-chain messenger RNA, demonstrated that this plasma-cell population was *lambda–light-chain restricted*, with a markedly skewed kappa-to-lambda ratio — the morphologic and immunophenotypic confirmation of a clonal, lambda-producing plasma-cell disorder. Congo red staining of the marrow core additionally showed amyloid deposition in vessel walls and the interstitium, again with apple-green birefringence under polarized light.

The case turns next on *typing the amyloid*, which is the single most consequential step in the pathological workup, because the systemic amyloidoses are morphologically indistinguishable on Congo red staining yet require entirely different treatments. We first attempted typing by immunohistochemistry and immunogold electron microscopy, applying antibodies against kappa and lambda light chains, transthyretin, and serum amyloid A protein. The lambda stain decorated the deposits more than the others, supporting AL (lambda) type. However, immunohistochemical typing of amyloid is well recognized to be unreliable: light-chain amyloid epitopes may be masked or altered within the fibril, background staining from circulating immunoglobulins is common, and antibody-based methods have limited sensitivity and specificity for amyloid typing in fixed tissue. For this reason, immunohistochemistry alone is insufficient to assign type with confidence.

Definitive typing was therefore performed by *laser-capture microdissection of the Congo red–positive deposits followed by liquid chromatography–tandem mass spectrometry (LC-MS/MS)*. The Congo red–positive regions were dissected from the tissue section under microscopic guidance, the captured material was digested into

peptides, and the peptides were identified by mass spectrometry against protein databases. This proteomic analysis identified, as the dominant species within the deposits, peptides corresponding to the *constant and variable regions of immunoglobulin lambda light chain*. It also identified the "amyloid signature" proteins that are present in all amyloid deposits regardless of type — serum amyloid P component, apolipoprotein E, apolipoprotein A-IV, and clusterin — which serves as an internal control confirming that genuine amyloid was sampled. Critically, transthyretin and serum amyloid A peptides were *not* enriched in the deposits, excluding ATTR and AA amyloidosis. The amyloid was thus typed unequivocally as *AL (lambda)*.

On electron microscopy, the deposits were composed of the rigid, non-branching, randomly oriented fibrils, 8 to 12 nm in diameter, that are characteristic of amyloid of any type.

The final pathological interpretation is systemic AL amyloidosis of lambda light-chain type, arising from a clonal lambda-restricted plasma-cell population in the marrow, with tissue (fat-pad and marrow) confirmation of amyloid by Congo red staining and apple-green birefringence and with the amyloid protein typed as immunoglobulin lambda light chain by mass spectrometry. The workup embeds an important teaching point: *demonstrating amyloid and typing amyloid are two separate tasks*, and because immunohistochemistry is unreliable, mass-spectrometry–based proteomic typing is the current reference standard. Assigning the type correctly is not an academic exercise — it determines whether the patient receives anti–plasma-cell therapy or a transthyretin-directed agent.

Discussion of Management

Dr. Anjali Desai (Hematology): Once tissue confirmed AL amyloidosis of lambda type, management proceeded along two simultaneous and interdependent tracks: hematologic therapy directed at the amyloidogenic plasma-cell clone, and supportive cardiac and systemic care. Prognosis in AL amyloidosis is driven above all by the extent of cardiac involvement.

We first established the cardiac stage, because risk stratification in AL amyloidosis rests on cardiac biomarkers. Using the widely applied biomarker staging system, which incorporates the N-terminal pro–B-type natriuretic peptide, the cardiac troponin, and the difference between involved and uninvolved free light chains (dFLC), this patient exceeded the threshold for all three markers: an NT-proBNP of 6,420 pg/mL, a high-sensitivity troponin T of 62 ng/L, and a dFLC of 229 mg/L. He therefore had *advanced-stage* cardiac AL amyloidosis, which carries a guarded prognosis and which shaped every subsequent decision, including transplant eligibility and the tolerability of therapy.

The central principle of treatment is that the amyloid fibrils are produced by a plasma-cell clone, so eliminating or deeply suppressing that clone — and thereby shutting off the supply of the toxic, amyloidogenic free light chain — is the only way to halt and potentially reverse organ damage. The depth and speed of the *hematologic response*, measured by the fall in the serum free light chains and the dFLC, are the key determinants of organ recovery and survival. The patient was treated with *daratumumab combined with bortezomib, cyclophosphamide, and dexamethasone* (the regimen abbreviated Dara-CyBorD). The addition of the anti-CD38 monoclonal antibody daratumumab to the bortezomib-based backbone has been shown to produce faster and deeper hematologic responses and improved organ-response rates in newly diagnosed AL amyloidosis, and it has become a standard frontline regimen. Bortezomib, a proteasome inhibitor, is particularly active against plasma cells, which are exquisitely dependent on proteasomal handling of immunoglobulin; corticosteroids and cyclophosphamide complete the regimen. Therapy in amyloidosis must be delivered cautiously, with attention to fluid balance during dexamethasone pulses and to the neuropathy, which can be aggravated by bortezomib; subcutaneous, once-weekly bortezomib dosing mitigates neurotoxicity.

We considered *autologous hematopoietic stem-cell transplantation*, which can produce durable responses in carefully selected patients with AL amyloidosis. However, eligibility is sharply limited by cardiac involvement: patients with advanced cardiac amyloidosis, markedly elevated natriuretic peptides, significant arrhythmia, or hypotension have a high transplant-related mortality and are generally not candidates. Given this patient's advanced cardiac stage, low blood pressure, and autonomic dysfunction, he was not an upfront transplant candidate; the goal was to achieve a deep hematologic response with chemoimmunotherapy and to reassess.

Supportive cardiac management in amyloid cardiomyopathy differs in important and counterintuitive ways from the management of ordinary heart failure with reduced ejection fraction, and getting it wrong can be harmful. The mainstay is careful management of volume with *loop diuretics*, titrated to relieve congestion while avoiding the over-diuresis that precipitates hypotension in a preload-dependent, stiff ventricle — a balance that was difficult in this patient, whose diuresis was repeatedly limited by orthostatic hypotension. Several drug classes that are beneficial in other forms of heart failure are *poorly tolerated or hazardous* in amyloid cardiomyopathy. Beta-blockers are often poorly tolerated because the amyloid ventricle depends on heart rate to maintain cardiac output across a fixed, low stroke volume, and because conduction disease is common. Non-dihydropyridine calcium-channel blockers (verapamil and diltiazem) and digoxin both bind avidly to amyloid fibrils and can produce exaggerated, sometimes dangerous effects, including profound bradycardia, heart block, and digoxin toxicity, and are generally avoided. Angiotensin-converting–enzyme inhibitors and angiotensin-receptor blockers frequently cause symptomatic hypotension in these patients, whose blood pressure is already low and whose autonomic reflexes are impaired — which, incidentally, is why this man's own antihypertensives had to be withdrawn before diagnosis.

The *autonomic and neurologic* complications required their own attention. Orthostatic hypotension was managed with nonpharmacologic measures (liberal salt intake where congestion permitted, compression stockings, slow positional changes, and elevation of the head of the bed) and, cautiously, with midodrine; fludrocortisone was used sparingly because of its tendency to worsen volume overload. The competing demands of decongesting a congested patient while supporting the blood pressure of an orthostatic one epitomize the therapeutic tightrope of cardiac amyloidosis. Because patients with amyloid atrial infiltration are at elevated risk of intracardiac thrombus and thromboembolism even in sinus rhythm when atrial mechanical function is severely impaired, the threshold for anticoagulation is low even in sinus rhythm, and it is mandatory if atrial fibrillation occurs.

The patient's clinical course reflected the seriousness of advanced cardiac AL amyloidosis. He tolerated the first cycles of Dara-CyBorD with careful supportive care, and over the ensuing months his serum free lambda level and dFLC fell substantially, indicating a hematologic response and suppression of the offending clone. His congestive symptoms stabilized with meticulous diuretic and blood-pressure management, although his functional limitation and orthostatic symptoms persisted, reflecting established organ infiltration that resolves only slowly, if at all, once the light-chain supply is controlled. The case underscores that, while modern therapy can arrest the disease and produce deep hematologic responses, the prognosis in AL amyloidosis is dominated by the degree of cardiac involvement at diagnosis, and that *earlier recognition* — before the heart is irreversibly infiltrated — is the most important therapeutic opportunity.

That brings me to the teaching points, which are the reason this case is worth presenting. First, *think of amyloidosis whenever wall thickness and electrocardiographic voltage are discordant* — a thick-walled ventricle with low or unimpressive voltage is infiltration until proven otherwise. Second, the *apical-sparing strain pattern* on echocardiography, diffuse subendocardial late enhancement with abnormal nulling and expanded extracellular volume on magnetic resonance imaging, and a disproportionately elevated natriuretic peptide with a persistently elevated troponin in the absence of coronary disease are all clues to cardiac amyloid. Third, *the systemic clues are often present years before the heart fails* and are easy to overlook: bilateral carpal tunnel syndrome, especially in an older man, is a recognized herald; periorbital purpura and macroglossia are highly suggestive of AL amyloidosis; and an unexplained sensory and autonomic neuropathy, nephrotic-range proteinuria, the loss of pre-existing hypertension, and an acquired bleeding tendency should all prompt the question. Fourth, and most important, *the diagnosis is not complete until the amyloid is typed.* Demonstrating amyloid by Congo red staining proves only that amyloid is present; because AL and ATTR amyloidosis have entirely different treatments — anti–plasma-cell chemotherapy for AL, transthyretin stabilizers or silencers for ATTR — and because a coincidental monoclonal gammopathy is common in the elderly, every patient must have the amyloid protein definitively typed, ideally by mass spectrometry, before therapy is chosen. A negative pyrophosphate scan in a patient with a monoclonal protein, as here, points away from ATTR and toward AL, but it is the typed tissue, not any single noninvasive test, that decides.

Anatomical / Final Diagnosis

Systemic AL (lambda immunoglobulin light-chain) amyloidosis with cardiac, renal, and soft-tissue involvement.